The 2026 Strait of Hormuz Closure: Crude Flows, Import Composition, and Retail Pass-Through in the United States and China

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1. RESEARCH QUESTION AND DESIGN

How did the 2026 Strait of Hormuz closure change crude oil flows and import composition for the US and China relative to a fixed pre-crisis baseline, and how differently did the shock pass through to retail fuel prices in each economy?

This comparison is motivated by an exposure asymmetry. The United States imports around 0.43 mb/d of Hormuz-transiting crude, whereas China imports around 5.4 mb/d through the same chokepoint (EIA, World Oil Transit Chokepoints, 1H2025). This implies an exposure ratio between China and the US of roughly 12:1. The two countries also differ institutionally. US pump prices are determined by market fluctuations, in contrast to China, where the NDRC administers prices under a published formula. Therefore, crude pass-through to pump prices is estimated econometrically for the US and measured as a policy wedge for China. These are different quantities and are never placed in the same regression.

Design constraints. Treatment is ongoing and multistage; there is no clean post-period. There is no control group, since oil is a single global market. The closure is confounded with the concurrent Iran war, Iran's own export collapse, and the US naval blockade of Iranian ports declared on 12 April. These cannot be separated using observational data alone. Every residual reported below is therefore a crisis-period effect rather than a closure-specific one.

Demand was deliberately excluded from the question, as it is not observable at this horizon. The original interest in the effect on a typical citizen was sharpened into retail price pass-through, which is observable in both countries.

Baseline. The baseline is fixed at 28 Feb 2025 – 27 Feb 2026 and is never rolling. Because most series are monthly, the window is rendered two ways — Mar-2025 to Feb-2026 and Feb-2025 to Jan-2026 — and every estimate is reported across both. The second window exists because February 2026 shows evidence of pre-positioning, and including an inflated month in the baseline would inflate every measured decline.

2. DATA AND VALIDATION

Seven processed datasets were built from EIA, JODI, GACC, FRED, NDRC and WTO DataLab sources. The organising principle is one file per observation frequency, and never a master table. Broadcasting monthly data onto daily rows invents observations, so joins are performed at the point of a specific question, which keeps the aggregation choice visible in the analysis rather than buried in the data.

Several cleaning decisions materially affect the results. JODI reports five units per observation, so failing to filter on `UNIT_MEASURE` quintuples every row, and `CONVBBL` is the tonne-to-barrel conversion factor rather than a quantity. GACC publishes Table 14 in three different column layouts across the thirteen files used, so rows are located by commodity rather than by position. EIA's import series places aggregate codes (`NUS-Z00`, `ME0`, `MN0`, `MP0`) alongside individual countries, so summing without excluding them triple-counts.

Validation. Four independent checks were passed. US country sums equal the `NUS-Z00` aggregate exactly in all 89 months. Chinese volumes computed from GACC tonnages agree with JODI in 10 of 12 overlapping months. The SPR pipeline reproduces the announced 2022 strategic release at −985.9 kb/d against a stated 1 mb/d, using no information about that policy. And the chain of announced NDRC price changes reconciles with the published Beijing price levels across all 25 adjustment windows.

3. DID THE DISRUPTION OCCUR?

Figure 1: Strait of Hormuz transits, Mar 2025 – Jul 2026. Shaded region is the fixed baseline; dashed line marks the 28 Feb 2026 closure.

Tanker calls and cargo volumes both fell by a factor of 10 to 12 against baseline. This was tested across three baseline definitions (equal-length, pre-spike, and fixed-year) because a pre-closure run-up in tanker calls — above 80 per day in mid-February against 30 to 40 in December and January — suggested the baseline itself might be contaminated by pre-positioning. The fixed-year baseline gives 12.07× and the other two give approximately 10×, so the result does not depend on the choice.

Two further observations support the interpretation. Container traffic collapsed alongside tanker traffic, which indicates the disruption was not tanker-specific. And average cargo per transit rose from a stable level near 47,000 tonnes per call to peaks above 120,000, meaning the transits that continued were substantially fuller. This is consistent with only escorted or high-value convoys crossing.

4. FLOW DISRUPTION

China

Seasonal factors were estimated on 2019–2025 only, so that the treatment period could not contaminate the profile. Each month was normalised by its own year's mean, which removes level differences between years, and the resulting ratios were collapsed by calendar month. Chinese refinery intake proves close to aseasonal, with a range of only 7 percentage points across the year and no spring turnaround signature. Crude imports are more seasonal, with an 11-point range. This matters because seasonality was the most obvious competing explanation for the observed decline, and it can now be ruled out on evidence rather than assertion.

2026	**Intake**	**Imports**
March	87 (0.5%)	254 (2.1%)
April	2,412 (14.9%)	2,281 (19.2%)
May	4,138 (25.6%)	4,057 (34.2%)

Table 1: Chinese seasonally adjusted shortfall below baseline, kb/d, with percentage of baseline in parentheses.

March is statistically indistinguishable from baseline. This is consistent with the three to four week Gulf-to-China shipping lag, since March arrivals were largely February loadings, and it means April is the first fully treated month. The escalation profile across the three months matches the event chronology.

The May import shortfall is approximately 75% of China's 5.4 mb/d Hormuz exposure. It is below the physical ceiling, as it must be, and close enough to it to be consistent with the loss of most Hormuz supply while non-Hormuz suppliers continued.

No inventory buffering. Crude entering Chinese refineries comes from imports plus domestic production, net of stock movements, so the gap between intake and imports isolates the latter two. That gap sat 24 and 12 kb/d above its baseline of 4,283 in April and May, which is under half a percent. Refiners cut runs in proportion to the import shortfall rather than drawing down inventories. Given that Chinese crude stocks are commonly estimated near one billion barrels, this appears to have been a choice rather than a constraint.

United States

Every monthly deviation in total US crude imports falls below 2σ against a month-over-month noise floor of 340 kb/d. This is not evidence that no effect occurred. Maximum Hormuz exposure is 430 kb/d, so a complete loss of every Hormuz barrel would register at only 1.26σ. In other words, the month-to-month variation in total US imports is comparable in size to the entire quantity at risk, so even the largest disruption physically possible would be hidden inside ordinary fluctuation. The series cannot answer the question, whatever the true effect

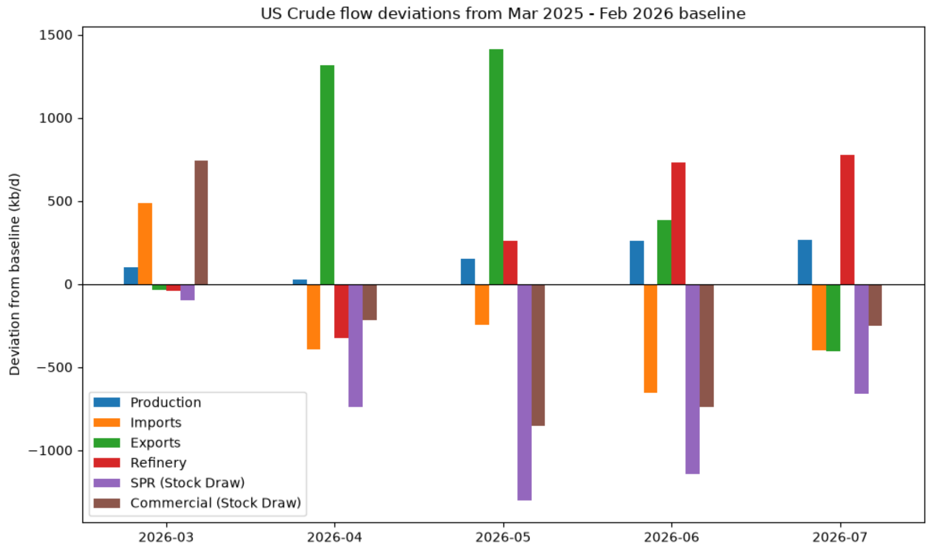

Figure 2: US crude flow deviations from the Mar 2025 – Feb 2026 baseline, kb/d. Not adjusted for seasonality.

turned out to be. This calculation was performed before any estimation, so the standard was set independently of the result.

The adjustment ran through other channels entirely. Measured against series-specific noise floors:

May 2026	**Deviation**	σ
Crude exports	+1,411	4.2
SPR draw	−1,300	>10
Commercial stock draw	−855	—
Crude imports	−245	0.7

Table 2: US crude flow deviations for May 2026 against series-specific month-over-month noise floors.

Exports rose by three to five times the amount by which imports fell. If the United States had been scrambling to cover a domestic shortfall, exports would have fallen. Instead, the strategic reserve was drawn at a rate exceeding the 2022 release while more crude was sent abroad. The US was not absorbing a supply shock so much as supplying a tight global market, with the reserve financing it.

5. IMPORT COMPOSITION

Disaggregating by origin resolves what the aggregate could not. Gulf-origin crude (Saudi Arabia, Iraq, Kuwait, UAE) fell 354 kb/d in May 2026, which is 2.3σ and a decline of roughly two thirds, and the result holds across both baseline windows. The Gulf-origin series is smaller and far less volatile than total imports, so the same disruption registers as a much larger multiple of that series' own noise: 2.3σ here against 0.7σ in the aggregate. Measuring the channel that was actually disrupted, rather than the total it sits inside, is what makes the effect visible.

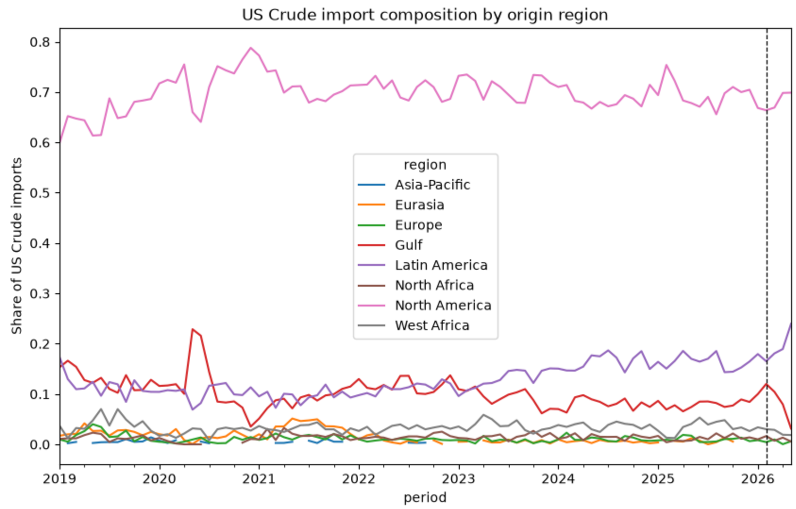

Figure 3: US crude import shares by origin region. Dashed line marks the closure.

Latin America absorbed the entire shortfall and more, rising 420 kb/d at 3.2σ, while every other region fell. North America was flat at 0.19σ despite supplying 4,248 kb/d at baseline. Canada is eight times the Gulf's volume with obvious spare capacity, and it did not step up.

This points to grade compatibility rather than volume availability as the binding constraint. Gulf Coast refineries are configured for medium-sour crude, and heavy-sour Latin American barrels are a closer substitute than light-sweet Canadian synthetic. It should be stressed that this is inferred from which regions moved, not from crude assay data; contract structure or freight economics could produce the same pattern.

February 2026 Gulf-origin imports ran 48% above baseline before collapsing. This is the same pre-positioning signature observed in mid-February tanker calls and in the March commercial stock build.

6. RETAIL PASS-THROUGH

United States

Augmented Dickey-Fuller tests fail to reject a unit root in levels for all four series and reject decisively in first differences, so estimation proceeds in differences. This is a necessary step rather than a formality: a regression between two $I(1)$ series in levels is spurious. Weekly changes in retail price are regressed on weekly changes in crude at lags 0 through 8, with HAC standard errors (maxlags = 8) to account for the serial correlation that overlapping lag windows induce. The specification follows Borenstein, Cameron and Gilbert (1997), simplified to a distributed lag. Daily crude is aggregated to Friday-ending weeks and matched to the following Monday's retail survey, so that no pump price is regressed on crude prices that had not yet been observed.

Cumulative pass-through is the sum of the nine estimated coefficients — lag 0, the contemporaneous week, through lag 8 — expressed in dollars per gallon for each dollar per barrel of crude. Complete mechanical pass-through is $1/42 = 0.0238$, since a barrel contains 42 gallons.

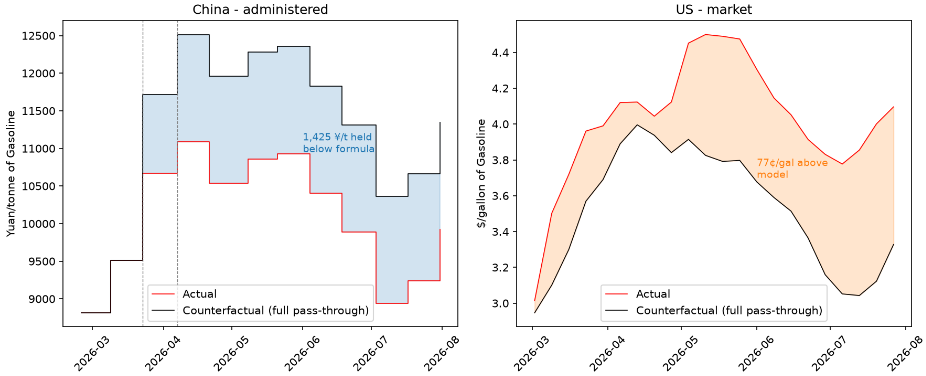

Figure 4: Gasoline, actual against counterfactual. Left: Beijing maximum retail price against the path implied by the NDRC formula. Right: US retail against a projection from a pre-crisis pass-through model.

Product	**Benchmark**	**Cumulative**	**s.e.**
Gasoline	Brent	0.0213	0.0029
Gasoline	WTI	0.0244	0.0038
Diesel	Brent	0.0253	0.0064
Diesel	WTI	0.0280	0.0068

Table 3: Cumulative US pass-through, \$/gal per \$/bbl, estimated on pre-crisis data. Complete mechanical pass-through is $1/42 = 0.0238$.

No t-statistic against the benchmark exceeds 1 in absolute value. Pre-crisis US pass-through is therefore statistically indistinguishable from complete, for both products and both benchmarks. Brent and WTI also do not differ meaningfully from one another, which means the near-tripling of the Brent-WTI spread during the crisis cannot be read off these coefficients.

Counterfactual. Rather than re-estimating on the crisis window, which would leave 22 observations against 10 parameters, the pre-crisis model is applied to crisis-period crude with nothing fitted on crisis data. Predicted weekly changes are cumulated from the last pre-crisis retail level to give a counterfactual price path.

By late July, gasoline sat 77 cents per gallon and diesel 97 cents per gallon above that path. However, the mean weekly residual is insignificant, with t between 0.94 and 1.42, and the three largest weeks account for essentially the entire gap while the remaining nineteen net to under half a cent. Those weeks are 9 March, 27 April and 4 May, each immediately following an escalation event: the IRGC's confirmation of the closure, the naval blockade, and the re-closure. In the week of 4 May, retail rose 32.9 cents while crude was falling.

This is a discrete-shock pattern rather than a change in pass-through regime. The natural mechanisms are refining margin expansion or a risk premium at the pump, neither of which can be tested without crack spread data.

China

The NDRC publishes an implied price change only when it chooses not to follow the formula. Substituting the actual change wherever no implied figure exists therefore yields a complete full-pass-through series, which is cumulated from the 3 February Beijing level to give a counterfactual path.

Across the 24 consecutive adjustment windows from February 2025 to January 2026, no narrowing notice was issued. Two were issued within six weeks of the closure.

Date	Implied	Actual	Absorbed
23 Mar 2026	2,205	1,160	1,045
7 Apr 2026	800	420	380

Table 4: NDRC interventions, yuan per tonne of gasoline. Implied figures are published by NDRC only where they deviate from the actual adjustment.

In total 1,425 yuan per tonne of gasoline increase was withheld, equivalent to $24.42 per barrel, and 1,375 yuan per tonne ($26.89/bbl) for diesel. The conversion uses 8.5 bbl/tonne for gasoline and 7.45 for diesel, reflecting their densities of roughly 0.74 and 0.845 t/m^3. The exchange rate is 6.865 CNY/USD, the DEXCHUS mean over March and April 2026, a period in which the rate moved only 1.4%. Beijing gasoline peaked at 10,930 against a counterfactual 11.4% higher, with diesel 12.1% higher.

The wedge steps twice and then holds perfectly flat across the remaining eight adjustments, including a 950 yuan per tonne reduction on 3 July. The constraint on prices was therefore active only while crude was rising steeply, and was let go once it fell back. This says something specific about *when* an administered pricing regime actually departs from its own formula, which a single average pass-through figure would average away entirely.

It is worth being precise about the legal basis. Article 6 of the Petroleum Price Management Measures already deducts the refiner processing margin progressively above $80 per barrel, and that deduction is contained within the implied figure NDRC publishes. The wedge is therefore Article 7 discretion in full, which requires State Council approval. Article 7 is also cited routinely for sub-50 yuan deferrals, so its invocation alone is not evidence of intervention; the narrowing of a large adjustment is.

7. SYNTHESIS

The exposure asymmetry did not translate into a proportional consumer outcome. China lost roughly a third of its crude imports and cut refinery runs in step. The United States lost two thirds of a channel worth 8% of its imports and replaced it within a month. Yet US pump prices rose 54% between February and May, from $2.91 to $4.48 for gasoline and $3.72 to $5.60 for diesel, while Beijing gasoline rose 26%. The country with twelve times the physical exposure experienced half the retail increase.

The explanation is institutional rather than physical. US pass-through was complete before the crisis and remained so through it, so the crude price reached consumers because nothing stood between them and it. China's formula would have delivered a comparable increase, and the state overrode it twice.

Products were hit harder than crude in both countries. Chinese refined product imports fell 48.6% year on year against 29.0% for crude, and in the US the diesel counterfactual gap of 97 cents exceeded gasoline's 77 cents, while in China diesel's absorbed wedge was larger per barrel than gasoline's. Two countries, independent datasets, and the same asymmetry. This is consistent with Hormuz crude being distillate-rich and with the Gulf being a major refined product exporter in its own right.

Each country used the buffer available to it. The US had exports, a strategic reserve, commercial inventories and grade-flexible substitutes, and drew on all four while allowing prices to clear. China had administered prices and used them, while refinery throughput absorbed the physical shortfall. Neither response is more effective in general; they are different margins of adjustment available to differently structured systems.

8. CAVEATS AND LIMITATIONS

Identification. There is no control group, and the closure cannot be separated from the war, Iran's export collapse, or the US blockade. Every residual reported is a crisis-period effect.

Statistical power. US crude import volume cannot detect this shock at any plausible effect size. The counterfactual test rests on 22 observations and cannot rule out a modest persistent shift in pass-through.

Asymmetric design. No free English-language source provides Chinese crude imports by origin at monthly frequency, so composition is analysed only for the US. Comtrade lacks Chinese monthly detail, and GACC Table 16 is value-only at HS 2-digit, where Chapter 27 mixes crude with coal, LNG and products, and omits Iraq, Kuwait and Angola. Separately, China reports close to zero crude from Iran while Malaysia shows $1.83bn of Chapter 27 trade, so transshipment makes any "imports from Iran" series unusable.

Origin is not route. Saudi and UAE barrels can reach open water via the Yanbu and Fujairah pipelines, so Gulf-origin US imports are an upper bound on Hormuz-transiting volume, and part of the observed decline may be rerouting rather than lost supply.

Measurement error correlates with treatment. AIS spoofing and dark-vessel activity increased during the closure, biasing transit counts in an unknown direction.

Specification. Working in first differences discards the long-run equilibrium relationship an error-correction model would capture, despite all four series being $I(1)$ at the same order. Asymmetry between rising and falling crude prices is assumed away, even though the window contains both. Kristoufek and Lunackova argue these series are fractionally integrated, in which case the standard $I(0)/I(1)$ dichotomy underlying the ADF test is misspecified; fractional integration is not estimable on a sample of this length.

Unverifiable inputs. The NDRC crude basket weighting is not published, so implied changes are taken as stated and cannot be independently reproduced. Chinese inventory data is unpublished. The Chinese pass-through fraction is endpoint-sensitive — 0.62 measured at the May peak and 0.47 at the end of July — because a fixed absolute wedge is divided by a moving denominator; the absolute figures are the stable ones.

Comparability. NDRC standard products are 89-octane gasoline and No. 0 diesel, which are not comparable in levels to US 87 AKI. Changes are comparable; levels are not.

Unresolved. A JODI-GACC discrepancy of roughly 14,500 kbbl in January and February 2026 offsets across the pair to 0.3% but remains undiagnosed, and February is the treatment-onset month. JODI reports Chinese refining at 16.4 mb/d for January 2026 against EIA's 14.2 for 2024, which appears definitional but is unresolved. Kiel Policy Brief 206 models this exact closure and has not yet been reviewed against these results.

SOURCES AND REFERENCES

Data

EIA API v2
US weekly petroleum supply (crude imports, exports, production, refinery inputs, commercial and SPR stocks) and monthly crude imports by country of origin. https://www.eia.gov/opendata/

EIA, World Oil Transit Chokepoints
Country Analysis Brief. https://www.eia.gov/international/content/analysis/special_topics/World_Oil_Transit_Chokepoints/ — 1H2025 Hormuz flows of 20.9 mb/d total oil, of which approximately 15 mb/d crude and condensate and 5.5 mb/d refined products, representing around 20% of global petroleum liquids consumption and one quarter of maritime traded oil. Exposure: China 5.4 mb/d against US 0.43 mb/d. This paper uses the 20.9 mb/d total-liquids figure for headline throughput.

FRED, St. Louis Fed
`DCOILBRENTEU` and `DCOILWTICO`, daily USD/bbl; `DEXCHUS`, daily CNY per USD. https://fred.stlouisfed.org/

JODI Oil World Database
Chinese monthly refinery intake (`REFINOBS`) and crude imports (`TOTIMPSB`) in KBD, with `CONVBBL` = 7.32 bbl/tonne. https://www.jodidata.org/

GACC
"(14) Major Import Commodities in Quantity and Value", monthly. Quantity in 10,000 tonnes, value in US$1,000. http://english.customs.gov.cn/statics/report/monthly.html — each month is published at its own UUID URL with no constructable pattern; per-file URLs are recorded in `docs/data_inventory.md`.

NDRC Price Department
Refined oil price adjustment notices, Jan 2025 – Jul 2026. Standard products are 89-octane gasoline and No. 0 vehicle diesel, priced in yuan per tonne as provincial and central-city maxima inclusive of consumption tax, VAT and surcharges, effective 24:00 on the adjustment date. https://www.ndrc.gov.cn/xwdt/ztzl/gncpyjg/ — per-notice URLs are recorded in `china-retail-prices-events.csv`.

WTO DataLab
Strait of Hormuz Trade Tracker; daily transits and cargo volumes. https://datalab.wto.org/Strait-of-Hormuz-Trade-Tracker

Policy and Regulation

Petroleum Price Management Measures (*Shiyou Jiage Guanli Banfa*)
NDRC, issued under Fagai Jiage [2016] No. 64, 13 January 2016. https://www.ndrc.gov.cn/xxgk/zcfb/tz/201601/t20160113_963566.html Article 6 sets the crude price bands: at or below $40/bbl prices are set as if crude were $40; between $40 and $80 a normal processing margin applies; above $80 the refiner processing margin is progressively deducted toward zero; at or above $130 prices are in principle not raised, or raised less, with accompanying fiscal measures. Article 7 sets a ten-working-day adjustment cycle effective at 24:00 on the publication date, provides that changes below 50 yuan/tonne are not made and carry into the next cycle, and permits the NDRC, with State Council approval, to suspend, delay or narrow an adjustment under special circumstances.

US Treasury OFAC
General Licence X, 22 June 2026, authorising Iranian oil sales through 21 August 2026. https://ofac.treasury.gov/recent-actions/20260622_33

Literature

Bacon, R. W. (1991)
"Rockets and Feathers: The Asymmetric Speed of Adjustment of UK Retail Gasoline Prices to Cost Changes." *Energy Economics* 13 (July).

Borenstein, S., Cameron, A. C. & Gilbert, R. (1997)
"Do Gasoline Prices Respond Asymmetrically to Crude Oil Price Changes?" *Quarterly Journal of Economics* 112(1): 305–339. doi:10.1162/003355397555118. https://faculty.haas.berkeley.edu/borenste/download/QJE97GasAsym.pdf — finds that nearly all of the response to a crude increase reaches the pump within four weeks while decreases pass through gradually over eight. Methodological basis for the US pass-through estimate here, simplified from an error-correction model to a distributed lag in first differences.

Kristoufek, L. & Lunackova, P.
"Rockets and Feathers Meet Joseph: Reinvestigating the Oil-Gasoline Asymmetry on the International Markets." arXiv:1407.5466 — argues these series are fractionally integrated, so the standard $I(0)/I(1)$ framework underlying error-correction specifications is misspecified, and finds no asymmetry in six of seven national markets. Cited here as a limitation; fractional integration is not estimable on this sample.

FTC, Bureau of Economics
"Gasoline Price Changes and the Petroleum Industry." Survey of asymmetric pass-through evidence.

Kiel Institute
Policy Brief 206, March 2026 — models a Strait of Hormuz closure scenario. Not yet reviewed against these results.

Chronology

UKOilWatch
Hormuz Timeline, 83 individually sourced events; primary index used to construct docs/chronology_table.xlsx. https://ukoilwatch.com/hormuz-timeline

Wikipedia
"2026 Strait of Hormuz crisis"; "2026 Strait of Hormuz campaign".

World Bank Data Blog
"Strait of Hormuz disruption sends oil prices surging."

News reporting
Per-event URLs (CNBC, CBS News, CNN, Al Jazeera, Al-Monitor, RFE/RL, gCaptain, Kpler via CNBC) are recorded in the Source column of chronology_table.xlsx. Most were transcribed from the UKOilWatch index rather than retrieved individually; live-update pages in particular are overwritten as stories develop and should be verified before quotation.

Conversion Factors

7.32 bbl/tonne for Chinese crude (JODI's own CONVBBL); 8.5 bbl/tonne for gasoline and 7.45 for diesel in the yuan-per-tonne to dollar-per-barrel conversion; 6.865 CNY/USD, the DEXCHUS mean over March–April 2026, over which the series ranged 1.4%; 42 US gallons per barrel.